

AI TRAINING – DAY 2 ASSIGNMENT

Python Fundamentals: Setup → Variables → Data Types → Input/Output → Type Conversion

Practical + MCQ + Debugging + Mini Challenge

Level	Beginner → Medium
Suggested Time	3–4 hours
Primary Goal	Understand Python basics by writing and fixing programs
Programs	10 required programs + debugging + challenge
Submission	One document/PDF + .py files or screenshots/output

1. Learning Goals

- Set up Python and verify it works.
- Understand variables and Python values.
- Understand int, float, str, bool and None.
- Understand input() and print().
- Understand why input() normally returns a string.
- Convert values between common data types.
- Write small programs independently.
- Read and fix simple Python errors.
- Explain every program in your own words.

2. Rules for This Assignment

- Type every program yourself; do not copy complete solutions.
- You may use AI to explain an error after attempting it yourself.
- For every program write: Input → Processing → Output.
- Test every program with at least 2 different inputs.
- Keep track of errors you faced and what you changed.

- Use Python 3.

3. Part A – Python Setup

1. Install Python 3 or use an online Python environment.

[Download Python](#)

[Python.org](https://www.python.org)

<https://www.python.org> › [downloads](#)

Using this I download

2. Verify the Python version from the terminal/command prompt.

By using “python --version” check the version.

3. Create a folder named Day02_Python_Basics.

Yes. I created folder with name DAY_02

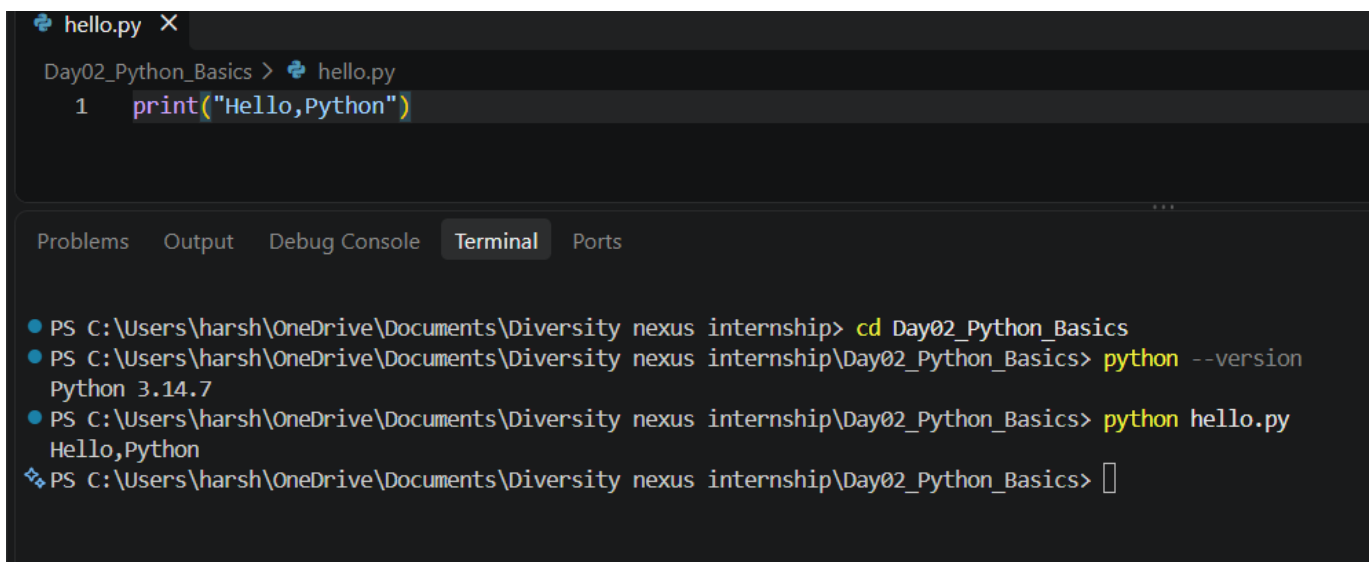
4. Create hello.py and print: Hello, Python!

5. Run the file successfully.

Yes run it successfully

6. Take a screenshot showing the version and successful output.

Expected evidence: Python version + hello.py + output.



```
hello.py X
Day02_Python_Basics > hello.py
1 print("Hello,Python")

Problems Output Debug Console Terminal Ports

PS C:\Users\harsh\OneDrive\Documents\Diversity nexus internship> cd Day02_Python_Basics
PS C:\Users\harsh\OneDrive\Documents\Diversity nexus internship\Day02_Python_Basics> python --version
Python 3.14.7
PS C:\Users\harsh\OneDrive\Documents\Diversity nexus internship\Day02_Python_Basics> python hello.py
Hello,Python
PS C:\Users\harsh\OneDrive\Documents\Diversity nexus internship\Day02_Python_Basics> 
```

4. Part B – Explore Before Coding

1. What does type(variable) tell you?

Variable is name which store the data.

2. create an int, float, str, bool and None value. Check their types.

```
age=21
per=78.1
name="pavi"
is_student=True
print(type(age))
print(type(per))
```

```
print(type(name))  
print(type(is_student))
```

Output:-

<class 'int'>

<class 'float'>

<class 'str'>

<class 'bool'>

3. What happens when you add an int and a float?

When you add int and float the output display in the float.because the float is larger range then int.

4. What happens when you join two strings with +?

```
name="pavi"  
surname="billupati"
```

Output :-

pavibillupati

5. What type does input() return when the user enters 25?

The integer value stores in the variable.

6. What happens with 10 + input('Enter number: ')? Why?

The 10 and the user send number both add

If the user send the string it gives the TypeError.

7. What is the difference between 10 and '10'?

10 is an integer and '10' is a string.

8. What is the difference between True and 'True'?

True is boolean and 'true' is a string.

9. What does int('25') produce?

The int() can covert the '25' int a integer.

10. What does int('25.5') do? Test it.

valueError:- because the

11. What does str(25) produce?

It produce '25'.

12. What do `bool(0)`, `bool(1)`, `bool("")` and `bool('hello')` produce?

False

True

False

True

13. What does `None` mean? Give one practical example.

5. Part C – Concept Understanding

Concept	Explain in your own words
Variable	Variable is name it can store the error ex:name="pavi"
int	It is datatype it say that the variable stores the whole numbers.ex:-a=12
float	It is datatype it say that the variable stores the integers numbers.ex:-b=10.0
str	it is datatype it say that the variable stores the words and sentences .ex:-name="pavithra".
bool	it is datatype it say that the variable stores the true and flase only .ex:-a=TRUE
None	None means is used to store the nothing in the variable.example:-value=None
input()	It used to take inputs fro the user.example:-name=input("enter your name:")
print()	It used to give the output to the user.ex:-print("radha")
Type conversion	It can change one datatype to another datatype.ex:-age=int(23.8)

6. Part D – MCQs: Basic Level

1. Which function displays output?

- A. `input()`
- B. `print()`✓
- C. `output()`
- D. `show()`

2. Which stores whole numbers?

- A. float
- B. str
- C. int✓
- D. bool

3. What is 10.5?

- A. int
- B. float✓
- C. str
- D. bool

4. Which is a string?

- A. 25
- B. 25.0
- C. True
- D. '25'✓

5. Which is a boolean?

- A. yes
- B. TRUE
- C. True✓
- D. 'True'

6. What is type(None)?

- A. null
- B. NoneType✓
- C. empty
- D. bool

7. Which gets user input?

- A. scan()
- B. read()
- C. input()✓
- D. get()

8. What does str(100) produce?

- A. int 100
- B. 100.0
- C. '100'✓
- D. True

9. What does int('20') produce?

- A. '20'
- B. 20✓
- C. 20.0
- D. Always error

10. Which is a valid variable name?

- A. 2name
- B. student-name

- C. student_name✓
- D. class

7. Part E – MCQs: Medium Level

1. What type is input('Age: ')?

- A. int
- B. float
- C. str✓
- D. bool

2. What happens in 10 + '5'?

- A. 15
- B. 105
- C. TypeError✓
- D. 10.5

3. What is int(12.9)?

- A. 13
- B. 12✓
- C. 12.9
- D. Error

4. What is float('10')?

- A. 10
- B. '10.0'
- C. 10.0✓
- D. Error

5. What is str(True)?

- A. True
- B. 'True'✓
- C. 1
- D. '1'

6. What is bool(0)?

- A. True
- B. False✓
- C. 0
- D. Error

7. What is 5 + 2.5?

- A. 7
- B. 7.5✓

- C. '7.5'
- D. Error

8. Correct integer input?

- A. `int(input())`✓
- B. `input(int())`
- C. `integer(input())`
- D. `input().int()`

9. What is x after `x=None`?

- A. 0
- B. False
- C. ''
- D. None✓

10. Which can fail if user enters abc?

- A. `int(input())`✓
- B. `str(input())`
- C. `input()`
- D. `print(input())`

8. Part F – MCQs: Thinking Level

1. Why does `age=input()` not work with `age+5`?

- A. input is slow
- B. input returns str✓
- C. + is unsupported
- D. age becomes None

2. Type conversion means:

- A. Changing a variable name
- B. Changing a value's data type✓
- C. Deleting a variable
- D. Printing a value

3. What is `int('10.5')`?

- A. 10
- B. 10.5
- C. `ValueError`✓
- D. `TypeError`

4. Which produces 25.0?

- A. `float('25')`✓
- B. `int('25')`

- C. str(25)
- D. bool(25)

5. None represents:

- A. zero
- B. False
- C. absence of a value✓
- D. empty string

6. x='5', y=2. Which gives 52?

- A. x+y
- B. int(x)+y
- C. x+str(y)✓
- D. float(x)+y

7. Marks for an average should normally be:

- A. str
- B. bool
- C. int or float✓
- D. None

8. price=input(); total=price*2 means:

- A. always numeric
- B. string multiplication may repeat text✓
- C. price becomes float
- D. Python converts automatically

9. Best first step for a TypeError?

- A. Delete code
- B. Ignore it
- C. Read the error and inspect types✓
- D. Restart PC

10. Which is true?

- A. Variables have fixed types forever
- B. Variables can refer to values of different types✓
- C. Python has no types
- D. input is automatically int

9. Part G – 10 Required Practical Programs

Program	Task	Topics	Level
1 – Personal Introduction	Ask name, age and city. Print a simple introduction.	str + int + input + print	Basic
2 – Basic Calculator	Ask two numbers. Print	int/float + arithmetic	Basic

	sum, difference, product and division.		
3 – Shopping Bill	Ask item name, quantity and price. Calculate total.	str + int + float + conversion	Basic
4 – Temperature Converter	Ask Celsius and convert to Fahrenheit.	float + arithmetic	Basic
5 – Student Marks	Ask 3 subject marks. Calculate total and average.	numbers + conversion	Basic → Medium
6 – Age Calculator	Ask age and show age after 5, 10 and 20 years.	int + arithmetic	Medium
7 – Type Inspector	Show what input() returns, then convert the value to int/float where possible.	str + type + conversion	Medium
8 – Bill Splitter	Ask bill, number of people and tip %. Calculate total and per-person amount.	float + int + conversion	Medium
9 – Data Type Challenge	Create int, float, str, bool and None. Print values/types. Convert selected values and explain.	All required types	Medium
10 – Mini Profile Generator	Ask name, age, height, student status and optional phone. Display values and types.	All topics	Medium

For EVERY program submit:

Tast:1

Code:

```
Name=input("Name-->")
Age=int(input("Age-->"))
City=input("City-->")
print("my name is ",Name)
print("I am ",Age,"years old)
Print("I am from ",City)
```

Output:

C:\Users\harsh\OneDrive\Documents\Diversity nexus internship\Day02_Python_Basics> python pratices.py

Name-->Pavithra

Age-->21

City-->Nellore

My name is Pavithra.

I am 21 years old.

I am from Nellore.

Task2:

Code:

```
number1=int(input("enter the number1 :"))
number2=int(input("enter the number :"))
sum_numbers=number1+number2
Difference=number1-number2
Product=number1*number2
Division=number1//number2
print(sum_numbers)
print(Difference)
print(Product)
print(Division)
```

Output:

enter the number1 :123

enter the number :234

357

-111

28782

0

Task3:

Code:

```
Item_name=input("enter the item name ")
quantity=int(input("Quantity:"))
price_of_one_item=float(input("enter the price of onr item :"))
total_bill=quantity*price_of_one_item
print(total_bill,"is total bill")
```

Output:

enter the item name pen

Quantity:10

enter the price of one item :5

50 is total bill

Task4:

Code:

```
temp=int(input("enter the temperature in °C:"))  
Fahrenheit=float(temp*9/5)+32  
print("the temperature in Fahrenheit :",Fahrenheit)
```

Output:

enter the temperature in °C:45

the temperature in Fahrenheit : 113.0

Task5:

Code:

```
markes1=int(input("enter the subject1 markes :"))  
markes2=int(input("enter the subject2 markes :"))  
markes3=int(input("enter the subject3 markes :"))  
total=markes1+markes2+markes3  
average=float(total/3)  
print("the sum of all subject markes",total)  
print("the average of markes",average)
```

Output:

enter the subject1 markes :23

enter the subject2 markes :56

enter the subject3 markes :87

the sum of all subject markes 166

the average of markes 55.333333333333336

Task6:

Code:

```
current_age=int(input("enter your current age :"))  
print("Age after 5 years :",current_age+5)  
print("Age after 10 years :",current_age+10)  
print("Age after 15 years :",current_age+15)
```

Output:

enter your current age :21

Age after 5 years : 26

Age after 10 years : 31

Age after 15 years : 36

Task7:

Code:

```
number=input("enter the number :")  
print(type(number))  
number1=int(number)  
number2=float(number)  
print(type(number1))  
print(number1)  
print(type(number2))  
print(number2)
```

Output:

Case1

enter the number :65

<class 'str'>

<class 'int'>

65

<class 'float'>

65.0

Case2:

enter the number :pavi

<class 'str'>

Traceback (most recent call last):

File "C:\Users\harsh\OneDrive\Documents\Diversity nexus internship\Day02_Python_Basics\practices.py", line 50, in
<module>

```
number1=int(number)
```

ValueError: invalid literal for int() with base 10: 'pavi'.because value Error occurred because "pavi" is text and cannot be converted into an integer using int()

Task8:

Code:

```
Bill=float(input("enter the Bill :"))
Tip_per=int(input("enter the tip percentage :"))
person=int(input("no.of persons :"))
Tip=Bill*Tip_per/100
Total=Bill+Tip
per_person=Total/person
print("the tip is ",Tip)
print("the total bill :",Total)
print("for each person the bill is ",per_person)
```

Output:

enter the Bill :678

enter the tip percentage :65

no. of persons :8

the tip is 440.7

the total bill : 1118.7

for each person the bill is 139.8375

Task9:

Code:

```
number = 25
decimal = 25.5
name = "Pavithra"
is_student = True
value = None
print(number)
print(type(number))
print(decimal)
print(type(decimal))
print(name)
print(type(name))
print(is_student)
print(type(is_student))
print(value)
print(type(value))
a = 25
print(str(a))
print(type(str(a)))
print(float(a))
print(type(float(a)))
b = "50"
print(int(b))
print(type(int(b)))
```

Output:

25

<class 'int'>

25.5

<class 'float'>

Pavithra

<class 'str'>

True

<class 'bool'>

None

<class 'NoneType'>

25

<class 'str'>

25.0

<class 'float'>

50

<class 'int'>

Task10:

Code:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
height = float(input("Enter your height: "))
student = bool(input("Enter student status: "))
phone = input("Enter your phone number(it is optional): ")
print("Name:", name, type(name))
print("Age:", age, type(age))
print("Height:", height, type(height))
print("Student:", student, type(student))
print("Phone:", phone, type(phone))
```

Output:

Enter your name: pavithra

Enter your age: 21

Enter your height: 134

Enter student status: yes

Enter your phone number(it is optional): 123456789

Name: pavithra <class 'str'>

Age: 21 <class 'int'>

Height: 134.0 <class 'float'>

Student: True <class 'bool'>

Phone: 123456789 <class 'str'>

10. Part H – Debugging Lab

Bug 1

```
age=input('Enter age: ')
print(age+5)
```

Explain and fix.

Corrected code / explanation

```
age=int(input("Enter age:"))
print(age+5)
```

Because input() can take the input from user like a string

Bug 2

```
price=input('Price: ')
quantity=input('Quantity: ')
print(price*quantity)
```

Explain and fix.

TypeError

Corrected code / explanation:

```
Price=int(input('price:'))
Quantity=int(input('Quantity:'))
```

Print(price*Quantity)

Because the price and quantity should be in integer or float

Bug 3

```
marks='90'  
print(marks+10)
```

Explain and fix.

Corrected code / explanation:

```
marks=int('90')  
print(marks+10)
```

Because the string can't concatenate with integer

Bug 4

```
temperature=float(input('Temperature: '))  
print('Temperature is '+temperature)
```

Explain and fix.

Corrected code / explanation:

```
temperature=float(input('Temperature: '))  
print('Temperature is ',temperature)
```

Because "temperature is" is string that can concatenate with the temperature integer

Bug 5

```
number=int(input('Number: '))  
print('Number: '+number)
```

Explain and fix.

Corrected code / explanation:

```
number=int(input('Number: '))  
print('Number: ',number)
```

Same explanation as above

Bug 6

```
value=int(input('Number: '))  
print(float(value))
```

Does this work? Explain why

It take the input from the user as integer and stores in value .and the value is change into the float then it prints float

Ex:- output

Number:-10

10.0

Corrected code / explanation

The code is correct

11. Part I – Medium Challenge: Student Result Calculator

- Ask student name.
- Ask marks in 5 subjects and convert them to numbers.
- Calculate total and average.
- Display name, total and average.
- Create result_status as a boolean showing the calculation is completed.
- Create remarks=None first, then assign a suitable message.
- Display the type of important variables.
- Test with at least 3 different mark sets.
- Bonus: support decimal marks using float.

```
student_name=str(input("Enter the student name:"))
markes1=int(input("enter the subject1 markes :"))
markes2=int(input("enter the subject2 markes :"))
markes3=int(input("enter the subject3 markes :"))
markes4=int(input("enter the subject4 markes :"))
markes5=int(input("enter the subject5 markes :"))
total=markes1+markes2+markes3+markes4+markes5
average=total//5
print(total)
print(average)
result_status=bool(input("completed:"))
print(result_status)
remarks=None
remarks="calculation completed"
print(type(markes1))
print(type(markes2))
print(type(markes3))
print(type(markes4))
print(type(markes5))
```

```
Print(type(total))
```

```
print(type(average))
```

3 different mark sets:

Set-1

Enter the student name:pavithra

enter the subject1 markes :45

enter the subject2 markes :5

enter the subject3 markes :67

enter the subject4 markes :87

enter the subject5 markes :32

236

47.0

completed:yes

True

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'float'>

Set-2

Enter the student name:Venky

enter the subject1 markes :78

enter the subject2 markes :98

enter the subject3 markes :78

enter the subject4 markes :67

enter the subject5 markes :90

411

82.0

completed:yes

True

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'float'>

Set-3

Enter the student name:Sowmya

enter the subject1 markes :90

enter the subject2 markes :96

enter the subject3 markes :78

enter the subject4 markes :87

enter the subject5 markes :45

396

79.0

completed:yes

True

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'int'>

<class 'float'>

12. Part J – Programmer Thinking

Why doesn't Python automatically convert '25' to 25?

Answer: Python treats '25' and 25 are two different data types.

Python not automatically change the datatype when the change is need you use the int(),float(),str() and bool()

When would you choose int instead of float?

Answer: int is used for the whole numbers but the float is used for the integers.

Why is 'False' different from False?

Answer:'False' is a string because it is in quotes but,False is a boolean .

Why is None useful?

Answer: it used to store no value in it.

What is the difference between a syntax error and a type-related error?

Answer: syntax-error is means you wrote it incorrectly

type-related error is means wrote correctly,but the data types don't work together

Which concept was easiest? Why?

Answer: variables, because I understand that clearly

Which was confusing and what experiment helped?

Answer: no

13. Self-Check

Skill	Yes	Need Practice	Notes
Run a Python file	☒	☐	
Create variables	☒	☐	
Identify int/float/str/bool/None	☒	☐	
Explain input()	☒	☐	
Use print()	☒	☐	
Perform type conversion	☒	☐	
Understand bool conversion	☒	☐	

Read a simple error	☒	☐	
Write a small program independently	☒	☐	
Explain your own code	☒	☐	

14. Submission Checklist

- Setup screenshot
- Exploration answers
- Concept table
- 30 MCQ answers
- 10 programs + 2 tests each
- Debugging corrections
- Medium challenge
- Reflection answers
- Self-check

15. Suggested Evaluation

Area	What is evaluated	Weight
Concepts + MCQs	Basic understanding	20%
10 Programs	Logic, input/output, conversion	35%
Debugging	Error identification and correction	15%
Medium Challenge	Combining concepts	20%
Explanation	Can explain own code	10%

Day 2 principle: Don't memorize Python. Run it, break it, understand the error, fix it, and explain why it works.